

Nuclear Physics

Single Correct Type Questions

1. The ratio of the density of oxygen nucleus ($^{16}_8\text{O}$) and helium nucleus (^4_2He) is [25 Jan, 2023 (Shift-I)]

(a) 4 : 1 (b) 8 : 1
(c) 1 : 1 (d) 2 : 1

2. Read the following statements:

- (A) Volume of the nucleus is directly proportional to the mass number.
(B) Volume of the nucleus is independent of mass number.
(C) Density of the nucleus is directly proportional to the mass number.
(D) Density of nucleus is directly proportional to the cube root of the mass number.
(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options. [29 July, 2022 (Shift-II)]

(a) (A) and (D) only. (b) (A) and (E) only.
(c) (B) and (E) only. (d) (A) and (C) only.

3. Mass numbers of two nuclei are in the ratio of 4 : 3. Their nuclear densities will be in the ratio of

[26 July, 2022 (Shift-II)]

(a) 4 : 3 (b) $\left(\frac{3}{4}\right)^{\frac{1}{3}}$
(c) 1 : 1 (d) $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

4. The radius R of a nucleus of mass number A can be estimated by the formula $R = (1.3 \times 10^{-15}) A^{1/3}$ m. It follows that the mass density of a nucleus is of the order of:

$$(M_{\text{prot}} = M_{\text{neut.}} \simeq 1.67 \times 10^{-27} \text{ kg})$$

[3 Sep, 2020 (Shift-II)]

(a) 10^3 kg m^{-3} (b) $10^{17} \text{ kg m}^{-3}$
(c) $10^{10} \text{ kg m}^{-3}$ (d) $10^{24} \text{ kg m}^{-3}$

5. For a nucleus $^A_Z X$ having mass number A and atomic number Z [8 April, 2023 (Shift-I)]

- (A) The surface energy per nucleon (b_s) = $a_1 A^{2/3}$
(B) The Coulomb contribution to the binding energy

$$b_c = -a_2 \frac{Z(Z-1)}{A^{4/3}}$$

- (C) The volume energy $b_v = a_3 A$
(D) Decrease in the binding energy is proportional to surface area.
(E) While estimating the surface energy, it is assumed that each nucleon interacts with 12 nucleons, (a_1 , a_2 and a_3 are constants)

Choose the most appropriate answer from the options given below:

(a) (C), (D) only (b) (B), (C), (E) only
(c) (A), (B), (C), (D) only (d) (B), (C) only

6. The mass of proton, neutron and helium nucleus are respectively 1.0073 u, 1.0087 u and 4.0015 u. The binding energy of helium nucleus is: [01 Feb, 2023 (Shift-I)]

(a) 14.2 MeV (b) 28.4 MeV
(c) 56.8 MeV (d) 7.1 MeV

7. $^{238}_{92}\text{A} \rightarrow ^{234}_{90}\text{B} + ^4_2\text{D} + \text{Q}$

In the given nuclear reaction, the approximate amount of energy released will be:

[Given, mass of $^{238}_{92}\text{A} = 238.05079 \times 931.5 \text{ MeV/c}^2$, mass

of $^{234}_{90}\text{B} = 234.04363 \times 931.5 \text{ MeV/c}^2$, mass of $^4_2\text{D} = 4.00260 \times 931.5 \text{ MeV/c}^2$] [13 April, 2023 (Shift-I)]

(a) 3.82 MeV
(b) 5.9 MeV
(c) 2.12 MeV
(d) 4.25 MeV

8. The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as:

[28 June, 2022 (Shift-I)]

- (a) $Q = K_p$ (b) $(K_p + Q) < 0$
(c) $Q < K_p$ (d) $(K_p + Q) > 0$

9. Given the masses of various atomic particles $m_p = 1.0072 u$, $m_n = 1.0087 u$, $m_e = 0.000548 u$, $m_{\bar{\nu}} = 0$, $m_d = 2.0141 u$, where $p \equiv$ proton, $n \equiv$ neutron, $e \equiv$ electron, $\bar{\nu} \equiv$ antineutrino and $d \equiv$ deuteron. Which of the following process is allowed by momentum and energy conservation?

[6 Sep, 2022 (Shift-II)]

- (a) $n + n \rightarrow$ deuterium atom
(electron bound to the nucleus)
(b) $n + p \rightarrow d + \gamma$
(c) $p \rightarrow n + e^+ + \bar{\nu}$
(d) $e^+ + e^- \rightarrow \gamma$

10. Find the binding energy per nucleon for $^{120}_{50}\text{Sn}$. Mass of proton $m_p = 1.00783 u$, mass of neutron $m_n = 1.00867 u$ and mass of tin nucleus $m_{\text{Sn}} = 119.902199 u$.

(Take $1 u = 931 \text{ MeV}$) [4 Sep, 2020 (Shift-II)]

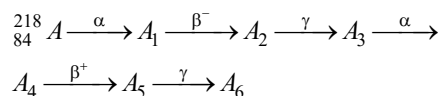
- (a) 9.0 MeV (b) 8.5 MeV
(c) 8.0 MeV (d) 7.5 MeV

11. Consider the nuclear fission $\text{Ne}^{20} \rightarrow 2\text{He}^4 + \text{C}^{12}$. Given that the binding energy / nucleon of Ne^{20} , He^4 and C^{12} are, respectively, 8.03 MeV, 7.07 MeV, and 7.86 MeV, identify the correct statement:

[10 Jan, 2019 (Shift-II)]

- (a) energy of 12.4 MeV will be supplied
(b) 8.3 MeV energy will be released
(c) energy of 9.72 MeV will be released
(d) energy of 11.9 MeV has to be supplied

12. Consider the following radioactive decay process

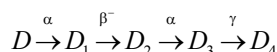


The mass number and the atomic number A_6 are given by:

[24 Jan, 2023 (Shift-I)]

- (a) 210 and 82 (b) 210 and 84
(c) 210 and 80 (d) 211 and 80

13. In the following nuclear reaction,



Mass number of D is 182 and atomic number is 74. Mass number and atomic number of D_4 respectively will be _____

[29 June, 2022 (Shift-II)]

- (a) 174 and 71 (b) 174 and 69
(c) 172 and 69 (d) 172 and 71

14. How many alpha and beta particles are emitted when Uranium ${}_{92}\text{U}^{238}$ decays to lead ${}_{82}\text{Pb}^{206}$?

[26 June, 2022 (Shift-I)]

- (a) 3 alpha particles and 5 beta particles
(b) 6 alpha particles and 4 beta particles
(c) 4 alpha particles and 5 beta particles
(d) 8 alpha particles and 6 beta particles

15. A nucleus of mass M emits γ -ray photon of frequency ' $h\nu$ '. The loss of internal energy by the nucleus is: [Take ' c ' as the speed of electromagnetic wave]

[20 July, 2021 (Shift-I)]

- (a) $h\nu \left[1 + \frac{h\nu}{2Mc^2} \right]$ (b) 0
(c) $h\nu \left[1 - \frac{h\nu}{2Mc^2} \right]$ (d) $h\nu$

Integer Type Questions

16. A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3 : 2. The ratio of their nuclear sizes will be $\left(\frac{x}{3} \right)^{\frac{1}{3}}$. The value of ' x ' is _____.

[25 Jan, 2023 (Shift-II)]

17. Assume that protons and neutrons have equal masses. Mass of a nucleon is $1.6 \times 10^{-27} \text{ kg}$ and radius of nucleus is $1.5 \times 10^{-15} A^{1/3} \text{ m}$. The approximate ratio of the nuclear density and water density is $n \times 10^{13}$. The value of n is _____

[24 Jan, 2023 (Shift-I)]

18. Nucleus A having $Z = 17$ and equal number of protons and neutrons has 1.2 MeV binding energy per nucleon. Another nucleus B of $Z = 12$ has total 26 nucleons and 1.8 MeV binding energy per nucleons.

The difference of binding energy of B and A will be _____ MeV .

[1 Feb, 2023 (Shift-II)]

19. A nucleus with mass number 242 and binding energy per nucleon as 7.6 MeV breaks into fragment each with mass number 121. If each fragment nucleus has binding energy per nucleon as 8.1 MeV , the total gain in binding energy is _____ MeV

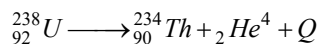
[8 April, 2023 (Shift-I)]

20. The minimum kinetic energy needed by an alpha particle to cause the nuclear reaction ${}^{16}_7\text{N} + {}^4_2\text{He} \rightarrow {}^1_1\text{H} + {}^{19}_8\text{O}$ in a laboratory frame is n (in MeV). Assume that ${}^{16}_7\text{N}$ is at rest in the laboratory frame. The masses of ${}^{16}_7\text{N}$, ${}^4_2\text{He}$, ${}^1_1\text{H}$ and ${}^{19}_8\text{O}$ can be taken to be $16.006 u$, $4.003 u$, $1.008 u$ and $19.003 u$, respectively, where $1 u = 930 \text{ MeVc}^{-2}$. The value of n is _____.

[JEE Adv, 2022]

21. From the given data, the amount of energy required to break the nucleus of aluminium ${}_{13}^{27}\text{Al}$ is _____ $\times 10^{-3} \text{ J}$.
[25 July, 2021 (Shift-II)]
 Mass of neutron = $1.00866u$
 Mass of proton = $1.00726u$
 Mass of Aluminium nucleus = $27.18846u$
 (Assume $1u$ corresponds to $x \text{ J}$ of energy)
 (Round off to the nearest integer)
22. A nucleus disintegrates into two nuclear parts, in such a way that ratio of their nuclear sizes is $1 : 2^{1/3}$. Their respective speed have a ratio of $n : 1$. The value of n is _____
[11 April, 2023 (Shift-II)]

23. A common example of alpha decay is _____
[12 April, 2023 (Shift-I)]



Given :

$${}_{92}^{238}\text{U} = 238.05060u$$

$${}_{90}^{234}\text{Th} = 234.04360u$$

$${}_2^4\text{He} = 4.00260u, \text{ and}$$

$$1u = 931.5 \frac{\text{MeV}}{c^2}$$

The energy released (Q) during the alpha decay of ${}_{92}^{238}\text{U}$ is _____ MeV



ANSWER KEY

- | | | | | | | | | | |
|----------|---------|---------|---------|---------|---------|----------|---------|-----------|------------|
| 1. (c) | 2. (b) | 3. (c) | 4. (b) | 5. (a) | 6. (b) | 7. (d) | 8. (d) | 9. (b) | 10. (b) |
| 11. (c) | 12. (c) | 13. (a) | 14. (d) | 15. (a) | 16. [2] | 17. [11] | 18. [6] | 19. [121] | 20. [2.32] |
| 21. [27] | 22. [2] | 23. [4] | | | | | | | |

EXPLANATIONS

1. (c) $\therefore$ Nuclear density $= \frac{Am}{\frac{4}{3}\pi R^3}$

Where A is the mass number and m is mass of one nucleon.

$$\therefore R = R_0 A^{\frac{1}{3}}$$

$$\Rightarrow R^3 = R_0^3 A$$

$$\Rightarrow \text{Nuclear density} = \frac{Am}{\frac{4}{3}\pi R_0^3 A}$$

$$\text{Nuclear density} = \frac{3m}{4\pi R_0^3}$$

$\therefore$ Nuclear density is independent of A

2. (b) Radius, $R = R_0 A^{\frac{1}{3}} \Rightarrow R \propto A^{\frac{1}{3}}$

$$V \propto R^3$$

$$\therefore V \propto A$$

And mass $\propto A$

So, density is independent of A .

3. (c) Radius of nucleus, $R = R_0 A^{\frac{1}{3}}$

$$\delta = \frac{\text{mass}}{\text{Volume}} = \frac{mA}{\frac{4}{3}\pi R^3} = \frac{mA}{\frac{4}{3}\pi R_0^3 A}$$

$\Rightarrow$ Density is independent of mass number

4. (b) Density of Nucleus $= \frac{\text{Mass}}{\text{Volume}}$

$$= \frac{1.67 \times 10^{-27} \times A}{\frac{4}{3}\pi (1.3)^3 \times 10^{-45} \times A} \approx 10^{17}$$

5. (a) $\therefore$ Mass number $A \propto r^3 \Rightarrow r \propto A^{\frac{1}{3}}$.

$$\therefore \text{Surface energy per nucleon} \propto \frac{r^2}{A} \propto \frac{1}{A^{\frac{1}{3}}}$$

A is incorrect

Coulomb contribution to the binding energy is

$$b_c = \frac{-a_2 Z(Z-1)}{A^{\frac{1}{3}}} \Rightarrow B \text{ is incorrect}$$

Volume energy $\propto A \Rightarrow C$ is correct

As we consider only surface energy contribution then option is correct. $\Rightarrow D$ is correct.

In case of surface energy, only 3 interactions contribute to surface energy. $\Rightarrow E$ is incorrect.

6. (b) B.E of Helium $= (2m_p + 2m_n - m_{He}) \times 931.5 \text{ MeV}$
 $= 28.4 \text{ MeV}$

7. (d) $Q = (m_A - m_B - m_D) \times c^2$
 $= (238.05079 - 234.04363 - 4.00260)$
 $= (0.00456 \text{ c}^2).u$

$$0.00456 \text{ c}^2 \cdot 931.5 \frac{\text{MeV}}{\text{c}^2} \quad \left[\because 1u = 931 \frac{\text{MeV}}{\text{c}^2} \right]$$

$$\Rightarrow 4.25 \text{ MeV}$$

8. (d) $A + p \rightarrow B + b$

$$Q = K_B + K_b - K_A$$

$$\Rightarrow Q + K_p = K_B + K_b$$

$$\Rightarrow Q + K_p > 0$$

9. (b) Δm should be positive

$$(m_p + m_n) > m_d$$

$\Rightarrow$ only (2) is possible

10. (b) $B.E. = (\Delta m)C^2$

$$\Delta m = (50 m_p + 70 m_n) - (m_{Sn})$$

$$= (50 \times 1.00783 + 70 \times 1.00867) - (119.902199)$$

$$B.E. = \Delta m \times 931 \text{ MeV}$$

$$\Rightarrow B.E. = 1020.5631 \text{ MeV}$$

$$\Rightarrow \frac{B.E.}{\text{Nucl.}} = \frac{1020.5631}{120} = 8.5 \text{ MeV} = 8.5 \text{ MeV}$$

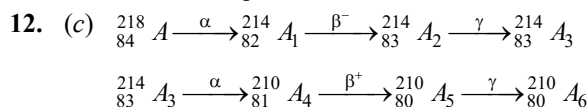
11. (c) Energy $= (\text{Binding Energy})_{\text{Reactant}} - (\text{Binding Energy})_{\text{Product}}$

$$= 20 \times 8.03 - (8 \times 7.07 + 12 \times 7.86)$$

$$= 160.6 - (56.56 + 94.32) = 160.6 - 150.88 = 9.72 \text{ MeV}$$

Hence, 9.72 MeV released. No option is correct.

Hence, this question is bonus.



13. (a) Let us suppose for D_4 ,

Atomic number = z , Mass number = A

$$\therefore A = 182 - 4 - 4 = 174$$

$$Z = 74 - 2 - 2 + 1 = 71$$

14. (d) When a radioactive nucleus decays by emitting an α -particle, its mass number decreases by 4 units and its atomic number decreases by 2 units.

$$\therefore \text{Number of } \alpha\text{-particles emitted} = \frac{32}{4} = 8.$$

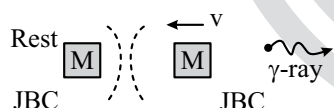
$$\text{Change in atomic number} = 92 - 16 = 76.$$

But the required atomic number is 82. This difference in atomic number is due to emission of β -particles. There is no change in mass number but its atomic number increases by 1 unit on the emission of a β -particle.

$$\therefore \text{Number of } \beta\text{-particles emitted} = 82 - 76 = 6.$$

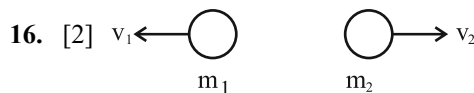
15. (a) Using conservation linear momentum, we can write

$$Mv = \frac{h}{\lambda} = \frac{hv}{c} \Rightarrow v = \frac{hv}{Mc}$$



Loss of internal energy = Gain in kinetic energy + energy of gamma ray

$$= \frac{1}{2}Mv^2 + hv = \frac{1}{2}M\left(\frac{hv}{Mc}\right)^2 + hv = hv\left[1 + \frac{hv}{2Mc^2}\right]$$



$$\text{Given, } \frac{v_1}{v_2} = \frac{3}{2}$$

Using conservation of linear momentum,

$$m_1v_1 = m_2v_2$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{2}{3}$$

Since, Nuclear mass density remains constant

$$\Rightarrow \frac{m_1}{\frac{4}{3}\pi r_1^3} = \frac{m_2}{\frac{4}{3}\pi r_2^3}$$

$$\Rightarrow \left(\frac{r_1}{r_2}\right)^3 = \frac{m_1}{m_2}$$

$$\Rightarrow \frac{r_1}{r_2} = \left(\frac{2}{3}\right)^{\frac{1}{3}}$$

$$\Rightarrow x = 2$$

$$\left[\because \frac{m_1}{m_2} = \frac{2}{3} \right]$$

17. [11] Density of nuclei = $\frac{\text{mass of nuclei}}{\text{volume of nuclei}}$

$$\rho = \frac{1.6 \times 10^{-27} A}{\frac{4}{3}\pi (1.5 \times 10^{-15})^3 A} = 0.113 \times 10^{18}$$

$$\rho_w = 10^3$$

$$\text{Hence } \frac{\rho}{\rho_w} = 11.31 \times 10^{13}$$

18. [6] For nucleus A mass number = 34

$$\text{Total binding energy} = 1.2 \times 34 = 40.8 \text{ MeV}$$

For nucleus B mass number = 26

$$\text{Total binding energy} = 1.8 \times 26 \text{ MeV}$$

$$= 46.8 \text{ MeV}$$

$$\text{Difference of BE} = 46.8 - 40.8 = 6 \text{ MeV}$$

19. [121] Initial binding energy = $242 \times 7.6 \text{ MeV}$

$$\text{Final binding energy} = (121 + 121) \times 8.1 \text{ MeV}$$

$$= 242 \times 8.1 \text{ MeV}$$

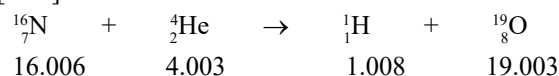
Total gain in binding energy = (Final binding energy – Initial binding energy)

$$= 242 (8.1 - 7.6)$$

$$= 242 \times 0.5$$

$$= 121 \text{ MeV}$$

20. [2.32]



$$16.006 \quad 4.003 \quad 1.008 \quad 19.003$$

$$4v_0 = 1v_1 + 19v_2 = 20v_2 \quad (\text{For max loss of KE})$$

$$v_2 = \frac{v_0}{5}$$

$$E \text{ required} = (1.008 + 19.003 - 16.006 - 4.003) \times 930 = 1.86$$

$$\frac{1}{2}4v_0^2 - \frac{1}{2}20v_2^2 = 1.86$$

$$\frac{1}{2}4v_0^2 - 10\frac{v_0^2}{25} = 1.86$$

$$2v_0^2 - \frac{2}{5}v_0^2 = 1.86, \Rightarrow \frac{8}{5}v_0^2 = 1.86$$

$$v_0^2 = \frac{1.86 \times 5}{8}$$

$$\text{KE} = \frac{1}{2}4v_0^2 = 2v_0^2 = \frac{1.86 \times 5}{4} = 2.325 \text{ MeV}$$

21. [27] Mass defect,

$$\Delta m = [Zm_p + (A - Z)m_n - M_{\text{Al}}]$$

$$= [13 \times 1.00726 + 14 \times 1.00866 - 27.18846]$$

$$= [(13.09438 + 14.12124) - 27.18846] u$$

$$= [27.21562 - 27.1884] u$$

$$= 0.02716 u$$

$$= 0.02716 \times J$$

$$= 27.16 \times 10^{-3}$$

22. [2]

23. [4] Mass defect, $\Delta m = (\sum m)_{\text{product}} - (\sum m)_{\text{reactant}}$

Energy released,

$$Q = \Delta m \cdot C^2 = (m_u - m_{\text{Th}} - m_{\mu e}) C^2 \cdot 931.5 \frac{\text{MeV}}{C^2}$$

$$= 0.0044 \times 931.5 \text{ MeV} = 4.0986 \text{ MeV}$$

